Chapter 1: Consistency and Consensus Models

1.1 Two-Phase Commit Protocol (2PC)

The Two-Phase Commit Protocol is a distributed consensus algorithm that ensures all database nodes commit or abort a transaction together.

Before understanding Two-Phase Commit Protocol, one must understand Distributed Transactions.

The protocol consists of a Prepare Phase and a Commit Phase coordinated by a master node.

Chapter 2: Replication and Quorum

2.1 Raft Consensus Algorithm

Raft is a leader-based consensus algorithm designed for state machine replication.

Raft Consensus Algorithm requires knowledge of Two-Phase Commit Protocol and leader election.

Nodes in Raft can be in Leader, Follower, or Candidate state.